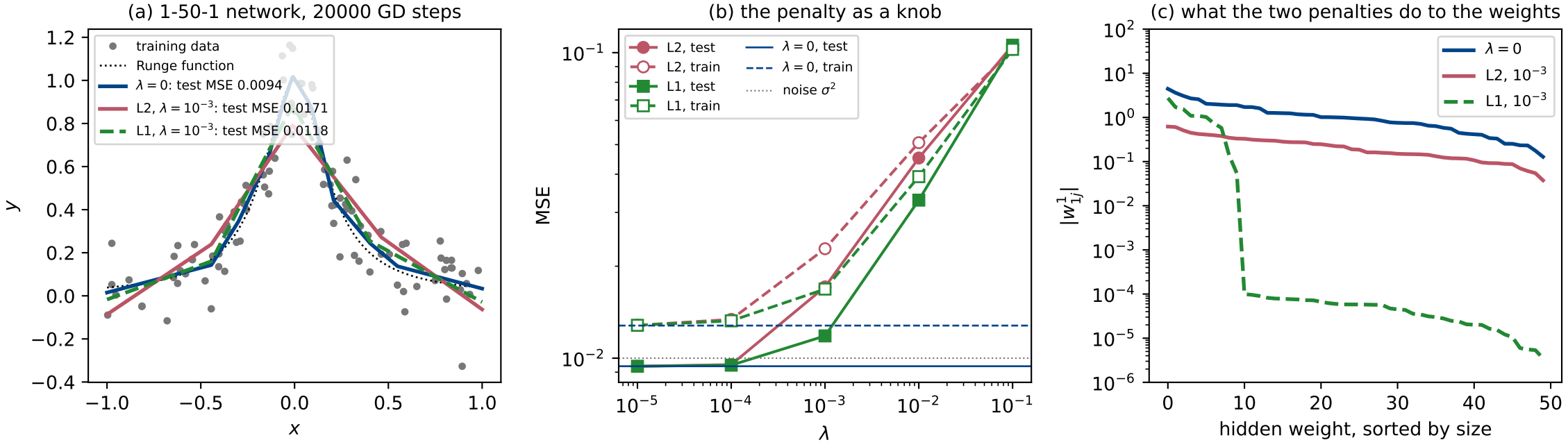
(a) 1-50-1 network, 20000 GD steps
training data
Runge function
$\lambda = 0$: test MSE 0.0094
L2, $\lambda = 10^{-3}$: test MSE 0.0171
L1, $\lambda = 10^{-3}$: test MSE 0.0118
y
x
(b) the penalty as a knob
L2, test
L2, train
L1, test
L1, train
$\lambda = 0$, test
$\lambda = 0$, train
noise σ^2
MSE
λ
(c) what the two penalties do to the weights
$\lambda = 0$
L2, 10^{-3}
L1, 10^{-3}
$|w^1_{1j}|$
hidden weight, sorted by size